

F1

$$(3) \quad R = \frac{D}{t}$$

$$D = L_f - L_o$$

$$D = 22791 \text{ km} - 22687 \text{ km}$$

$$D = 104 \text{ km} \quad t = 4 \text{ h}$$

$$\therefore R = \frac{D}{t} = \frac{104 \text{ km}}{4 \text{ h}} = 26 \text{ km/h} //$$

$$R = 26 \text{ km/h} \div 3,6 \cong 7,2 \text{ m/s} //$$

F2

(4) Sist No Conservativo

$$\Delta E_M = W_{Nc}$$

$$\Delta E_M = 0 - E_{PA}$$

$$\Delta E_M = -mgh_A \dots (1)$$

$$W_{Nc} = f_k d \cos 180^\circ$$

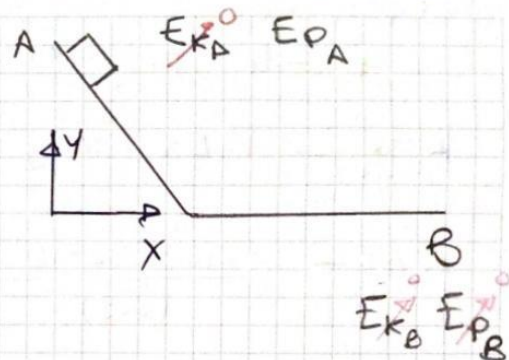
$$W_{Nc} = -\mu_k mg d \dots (2)$$

Remplazando

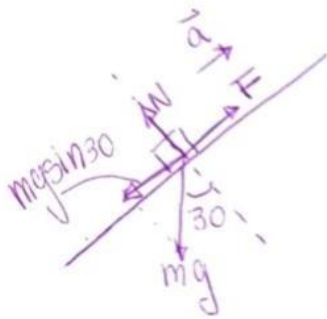
$$-\mu_k mg d = -mgh_A$$

$$d = \frac{h_A}{\mu_k} = \frac{2}{0.5}$$

$$d = 4 \text{ m}$$



F3



$$\Sigma F_{x'} = ma$$

$$F - mg \sin 30 = ma$$

$$150 - 10 \cdot 10 \cdot \frac{1}{2} = 10a$$

$$100 = 10a$$

$$\underline{a = 10 \text{ m/s}^2}$$

$$d = \frac{1}{2} a t^2$$

$$10 = \frac{1}{2} \cdot 10 t^2$$

$$\underline{t = \sqrt{2} \text{ s}}$$

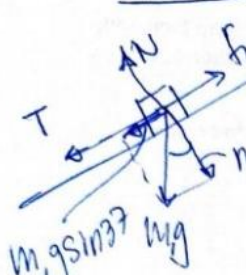
$$\Rightarrow P = \frac{W}{t} = \frac{F d}{t} = \frac{150 \cdot 10}{\sqrt{2}} \cdot \frac{\sqrt{2}}{\sqrt{2}}$$

$$\boxed{P = 750 \sqrt{2} \text{ W}} \quad \text{Sol.}$$

F4

P-1

D.C.L 8kg



$$\sum F_y = 0$$

$$N = m_1 g \cos 37$$

$$N = 8 \cdot 10 \cdot \frac{4}{5}$$

$$N_1 = 64 \text{ N}$$

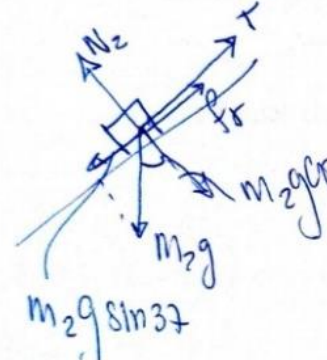
$$\sum F_x = m_1 a$$

$$T + m_1 g \sin 37 - f_r = m_1 a$$

$$T + 8 \cdot 10 \cdot \frac{3}{5} - \frac{1}{4} \cdot 64 = 8a$$

$$T + 32 = 8a \dots (1)$$

D.C.L. 4kg



$$\sum F_y = 0$$

$$N_2 = m_2 g \cos 37$$

$$N_2 = 4 \cdot 10 \cdot \frac{4}{5}$$

$$N_2 = 32 \text{ N}$$

$$\sum F_x = m_2 a$$

$$m_2 g \sin 37 - T - f_r = m_2 a$$

$$4 \cdot 10 \cdot \frac{3}{5} - T - \frac{1}{4} \cdot 32 = 4a$$

$$-T + 16 = 4a \dots (2)$$

Eq. (1) + (2)

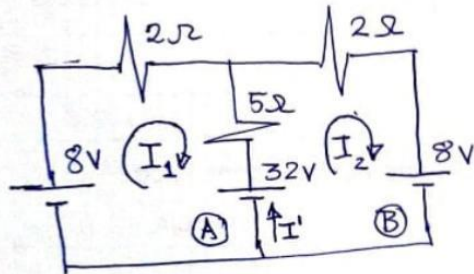
$$T + 32 = 8a$$

$$-T + 16 = 4a$$

$$48 = 12a$$

$$\boxed{a = 4 \text{ m/s}^2} \quad \text{Sol}$$

F5



$$\begin{cases} -7I_1 + 5I_2 = 24 \\ 5I_1 - 7I_2 = -24 \end{cases}$$

Resolviendo el
Sist. de Ecuaciones

$$I_1 = -2A \Rightarrow I_1 = 2A \uparrow$$

$$I_2 = 2A \uparrow$$

Malla A $\sum V_i = 0$

$$8 - 32 - 2I_1 - 5(I_1 - I_2) = 0$$

$$-7I_1 + 5I_2 = 24 \dots (1)$$

Malla B $\sum V_i = 0$

$$32 - 8 - 2I_2 - 5(I_2 - I_1) = 0$$

$$5I_1 - 7I_2 = -24 \dots (2)$$

$$I' = I_1 + I_2 \Rightarrow I' = 4A$$

$$P = (I')^2 R = 4^2 \cdot 5$$

$$\boxed{P = 80W}$$

Sol.